

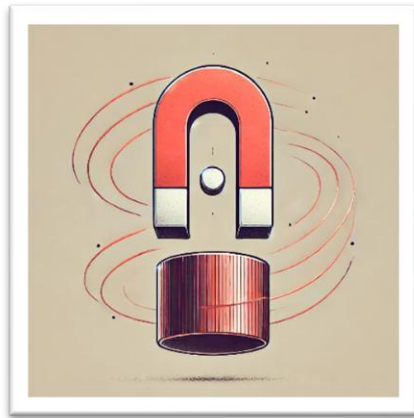
TO DEMONSTRATE
LENZ'S LAW USING A
FALLING MAGNET
THROUGH A COPPER
TUBE

Contents

S.no	Topic
1.	Aim
2.	Apparatus Required
3.	Theory
4.	Procedure
5.	Diagram
6.	Observation
7.	Result
8.	Conclusion
9.	Precautions
10.	Bibliography

Aim

To demonstrate Lenz's Law by observing the opposing force generated when a magnet falls through a copper tube and to study how the speed of the falling magnet is affected by the induced current.



Apparatus Required

- Copper Tube
- Aluminium Tube
- Plastic Tube
- Strong Magnet
- Non-Magnetic Rod (Plastic rod)
- Thread or String
- Timer
- Ruler or Measuring Tape
- Clamp Stand
- Marker or Tape

Theory

Lenz's Law states that the direction of an induced current in a conductor is such that the magnetic field created by this current opposes the change in magnetic flux that caused it. This principle ensures the conservation of energy, as the induced current resists the motion or change causing it.

In general, Lenz's Law is a fundamental concept in electromagnetic induction, explaining how currents are generated in conductors when exposed to a changing magnetic field. It plays a critical role in technologies like electric generators, transformers, and electromagnetic braking systems.



In this project, Lenz's Law is demonstrated using a magnet falling through a copper tube. As the magnet descends, its movement changes the magnetic flux through the tube, inducing eddy currents in the copper. These currents generate an opposing magnetic field that

slows the magnet's fall, providing a clear and tangible example of Lenz's Law in action.

Formula:

Lenz's Law is reflected in the formula of Faraday's law.

$$\varepsilon = - \frac{d\varphi}{dt}$$

where,

- ε - is the induced electromotive force (emf) in the conductor (the copper tube).
- φ - is the magnetic flux through the tube, given by $\varphi = \mathbf{B} \cdot \mathbf{A}$, where $\mathbf{B}$ is the magnetic field and $\mathbf{A}$ is the area through which the field lines pass.
- $\frac{d\varphi}{dt}$ - is the rate of change of magnetic flux, which in this case is due to the falling magnet.

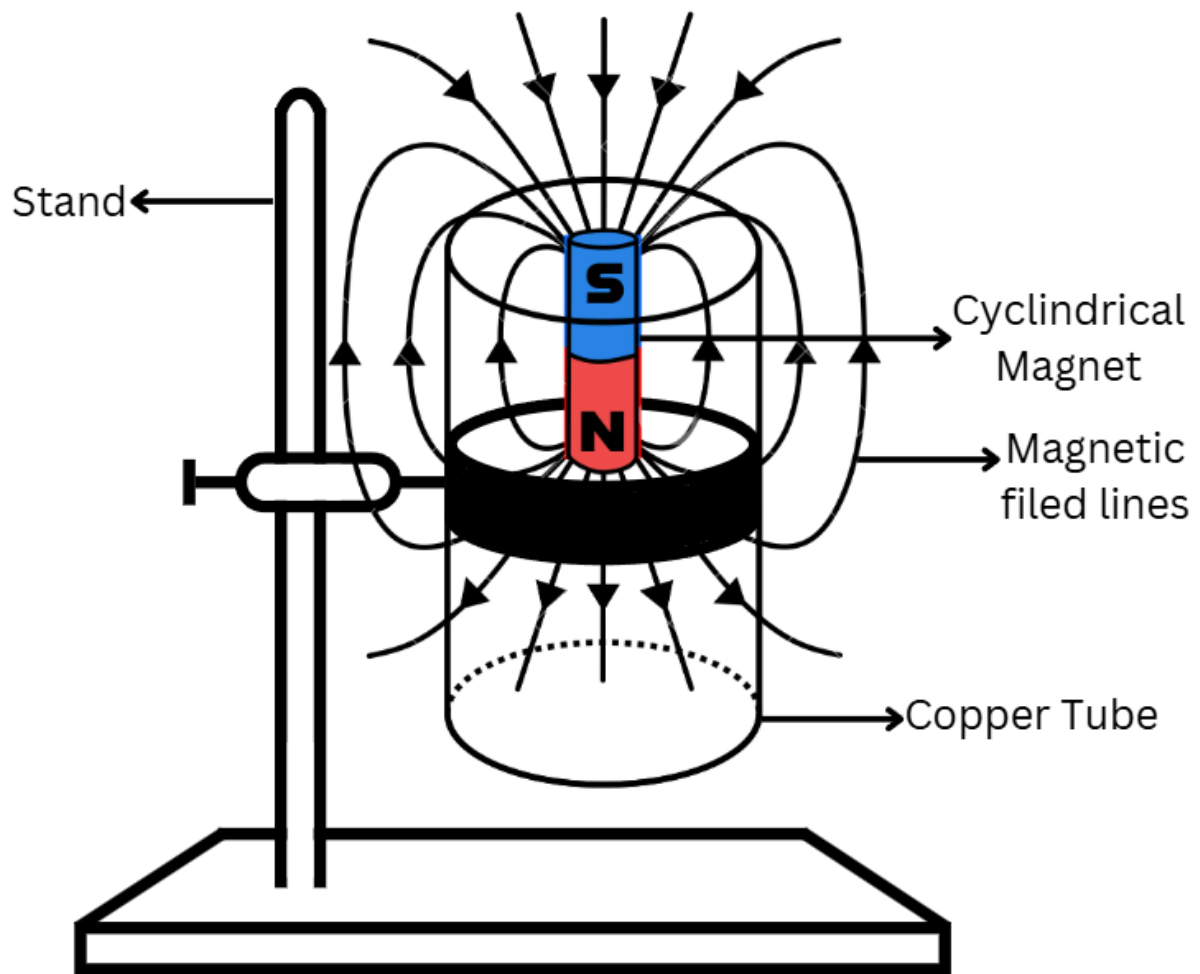
The negative sign indicates that the direction of the induced current is such that it opposes the change in the magnetic flux, as stated by Lenz's Law.

In the context of the project, the induced current leads to a resistive force on the falling magnet, which can be calculated using concepts from electromagnetic theory, though the exact calculation of the magnet's fall would depend on more detailed factors like the magnet's velocity, the resistance of the copper, and the strength of the magnetic field.

Procedure

1. Set up the copper tube vertically using a clamp stand.
2. Drop a non-magnetic object through the tube to observe its motion as a baseline.
3. Release the magnet at the top of the copper tube and allow it to fall.
4. Observe how the magnet's fall slows down due to the opposing force created by the induced currents in the copper tube.
5. Measure the time it takes for the magnet to fall through the tube.
6. Repeat the experiment with different tubes to explore variations.
7. Analyse the results, connecting them to Lenz's Law and the principle of conservation of energy.

Diagram



Observation

1. Without the Magnet:

- When a non-magnetic object (e.g., plastic rod) is dropped through the copper tube, it falls freely under gravity, with no noticeable resistance.

- The fall is smooth, and the object reaches the bottom of the tube in a short period.

2. With Magnet:

- Upon releasing the magnet from the top of the copper tube, the magnet falls more slowly compared to a non-magnetic object.
- As the magnet descends, it generates a changing magnetic flux through the copper tube, which induces eddy currents. These eddy currents create a magnetic field opposing the motion of the magnet, slowing it down.
- The magnet does not fall freely as it would through air. Instead, it experiences a noticeable resistive force, demonstrating the effects of Lenz's Law.

Object/Magnet	Fall time (seconds)	Speed of Fall (m/s)	Effect of Eddy Currents
Non-Magnetic Object	0.5	2.0	No resistance, falls freely
Magnet (Copper Tube)	1.2	0.83	Eddy currents induce opposing force, slows down significantly
Magnet (Aluminium Tube)	1.0	1.0	Eddy currents induce opposing force, weaker than copper
Magnet (Plastic Tube)	0.5	2.0	No resistance, falls freely

Result

- The fall time for the magnet through the copper tube is significantly longer (approximately 1.2 seconds) compared to the fall time of a non-magnetic object (approximately 0.5 seconds), demonstrating the resistance to the magnet's motion caused by induced currents.
- When the same experiment is conducted using an aluminium tube, the magnet still experiences resistance, but to a lesser extent than in the copper tube. This is because aluminium, while conductive, is not as effective as copper in generating eddy currents.
- In the plastic tube, the magnet falls at the same speed as a non-magnetic object, as there are no eddy currents generated due to the non-conductive nature of plastic.

Conclusion

The experiment demonstrates Lenz's Law, where the magnet's fall is slowed due to the eddy currents induced in the copper tube, which generate a magnetic field opposing the magnet's motion and it is confirmed that the induced currents oppose changes in magnetic

flux, illustrating energy conservation in electromagnetic induction.

Precautions

- Ensure the copper tube is free of any obstructions: Any irregularities or blockages inside the tube could interfere with the smooth fall of the magnet and affect the results.
- Use a strong magnet: A weak magnet may not generate sufficient magnetic flux change to induce noticeable eddy currents, reducing the observable effect.
- Keep the tube vertically aligned: The copper tube should be positioned vertically to ensure that the magnet falls straight down, as any tilt may alter the motion and slow the magnet unevenly.

Bibliography

- <http://hyperphysics.phy-astr.gsu.edu>
- <https://www.physicsclassroom.com>
- Halliday, D., Resnick, R., & Walker, J. (2013).
Fundamentals of Physics

